

STCLOUD 05.26

Hypervisor

Also known as a virtual machine monitor or VMM, is software that **creates** and **runs virtual machines**

A hypervisor allows a host computer to support multiple guest VMs by virtually **sharing its resources** (i.e. memory)

Hypervisors provide **greater IT mobility** since the guest VMs are independent of the host hardware

Multiple virtual machines can **run on one physical server**, **reducing space, energy, and maintenance** requirements

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<https://www.vmware.com/topics/hypervisor>; Images from Google Images

Benefits of Hypervisors

Speed

virtual machines can be **created instantly**, making it easier to provision resources as needed for **dynamic workloads**

Efficiency

running **several VMs on a physical machine is more efficient** than to run multiple underutilized physical machines

Flexibility

separates the OS from the underlying hardware, so the software no longer relies on specific hardware devices or drivers

Portability

multiple OS to reside on the **same physical server**; VMs in the hypervisor run **independently** from the physical machine; allows

Types of Hypervisors

Type 1 – Bare Metal or Native Hypervisor

- **Lightweight OS** directly running on top of the host machine
- **Isolated** from the guest OS, results in **better security**
- **Better performance** than hosted hypervisors
- Mostly used by **enterprise** for **data center** computing needs

Type 2 – Hosted or Client Hypervisor

- Runs as a **software** on an OS, like other programs
- Can still run the **same** or a **different guest OS** from the host OS
- Have **higher latency** than bare metal hypervisors